4 The planet Saturn has the most extensive and complex ring system of any planet in the Solar System. The rings have varying width and thickness. They are made up of mainly ice particles, with a trace of rocky material which are in orbit around Saturn. One of the rings, the D-ring, has an inner radius of 6.7×10^7 m.

(a) An object in krono-stationary orbit about Saturn has the same orbital period as the rotational period of Saturn. The rotational period of Saturn is 10 hours and 14 minutes.

A particle on the inner circumference of the D-ring has a speed of 2.38×10^4 m s ^{1}.

Deduce whether the particle is in krono-stationary orbit about Saturn.

.....

..... [2]

(b) By considering the orbit of the particle in **(a)** show that the mass of Saturn is $5.7 \cdot 10^{26}$ kg.

[3]

(c) Saturn's largest moon, Titan, is the second largest moon in the solar system.

Titan is 1.2×10^6 km from Saturn. The mass of Titan is 1.4×10^{23} kg.

Determine the distance from the centre of Titan where the resultant gravitational field strength between Titan and Saturn is zero.

distance = ... m [3]

[Total: 8]